PYTHON LAB FOR MECHANICAL APPLICATIONS

II B TECH I Semester : MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6ME15	ESC	0	0	2	1	40	60	100

COURSE OVERVIEW:

Python is a general-purpose, object-oriented programming language with high-level programming capabilities. It has apparent and easily understandable syntax, portability, and easy to learn.

COURSE OUTCOMES:

The course should enable the students to

1. Apply conditional statement, loops condition and functions in python program
2. Solve mathematical and mechanical problems using python program
3. Plot various type of chart using python program
4. Analyse the mechanical problem using python program
5. Illustrate programs on various python libraries such as numpy, pandas and matplotlib.

List of Python Programs

1. Write a program to find roots of quadratic equation.
2. Write a program to perform equations of uniform motion of kinematics:
   i. $v = u + at$
   ii. $s = ut + ½(at^2)$
   iii. $v^2 = u^2 – 2as$
3. Write a menu driven program to perform following properties of thermodynamics as given below:
   i. First Law of thermodynamics (U = Q - W), where ΔU is the change in the internal energy. Q is the heat added to the system, and W is the work done by the system.
   ii. Efficiency of Heat Engine = (TH – TC) / TH where TH & TC is the temperature of HOT and COLD Reservoirs.
4. Write the menu program to find the relationship between stress and strain curve as given below:
   i. E = Stress/Strain
   ii. G = Shear Stress/ Shear strain
   iii. Realation between E&C, E& K and E,C, and K
5. Write the program to determine the shear force and bending moment in beams.
   i. Cantlever beam Carrying Point Load at Center
   ii. Simply Supported beam Carrying UDL
6. Write a Program to find out unknown magnitude of T_B and T_D of unknown tensions can be obtained from two scalar equations of equilibrium i.e. ΣFx = 0 and ΣFy =0.
7. Write a program to perform interpolation of equally and unequally spaced data.
8. Write a program to calculate total pressure exerted in ideal fluid as equation is given below:
   P+1/2(ρv2) + ρgh = constant
   Where P is Pressure, V is Velocity of fluid, ρ is density and h is the height of the container.
9. Write a program to find and delete repeating number in Given List.
10. Write a program to input and print the element sum of user defined matrix.
11. Write a program to input and multiply two different matrices.
12. Write a program to compute eigen value and vector of a given 3*3 matrix using NumPy.
13. Write a program to draw line using equation y=mx+c
14. Write the program to determine the intersection point of two lines.
15. Draw various types of charts using matplotlib.
16. Write a program for Brasenham’s line drawing algorithm.
17. Write a program for geometric transformation of a given object.
18. Write program to perform the following operations on Strings.
    i. Concatenation
    ii. String Reverse